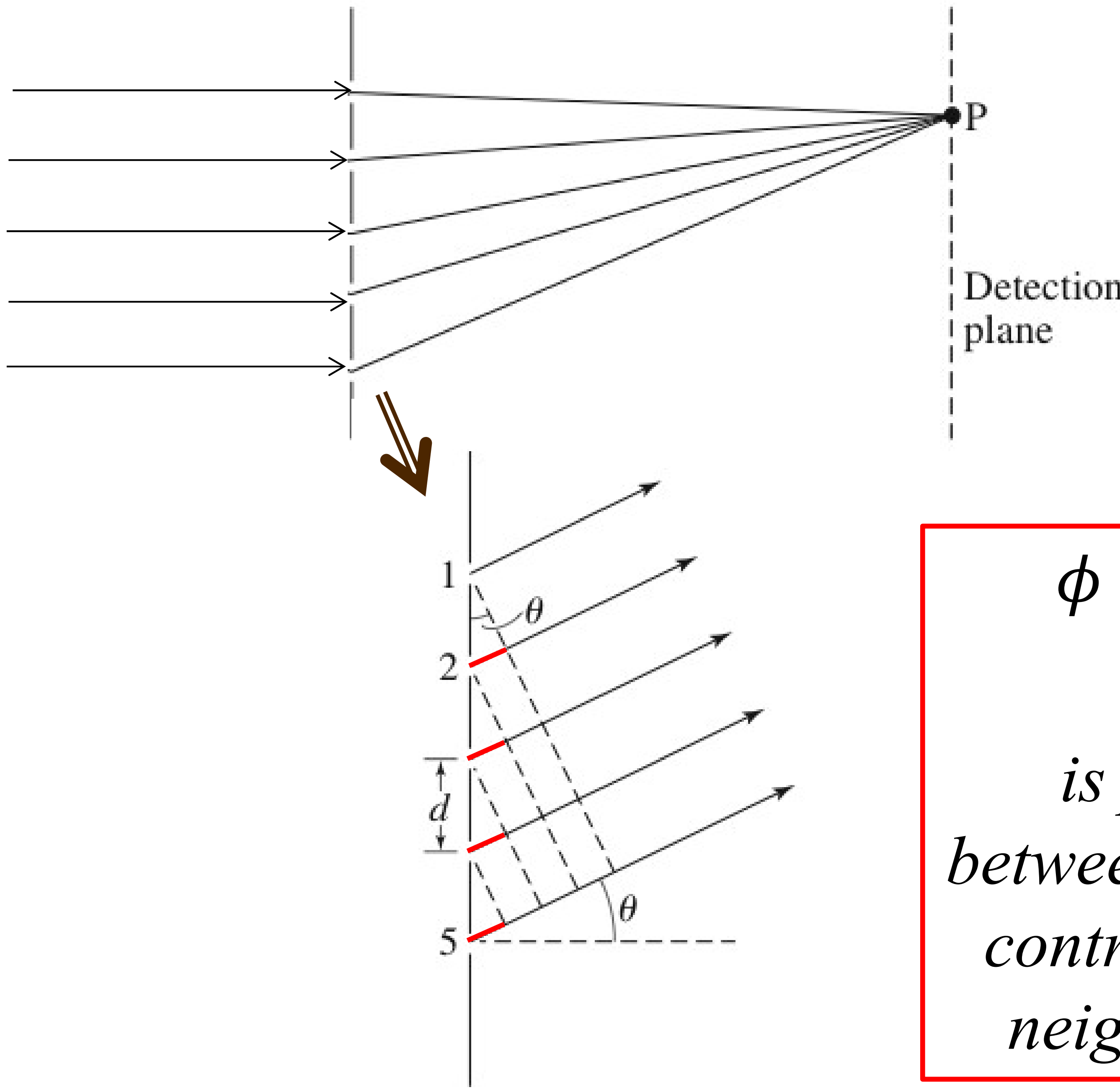
Ph134, Lecture 18

- **Fraunhofer diffraction:**
  - **Resolving power of diffraction gratings**
  - **Phasor addition approach to analyzing interference**

- **Fresnel diffraction:**
  - **Qualitative difference from Fraunhofer**
  - **Mathematical formalism**

Fraunhofer multiple-slit interference (diffraction grating):

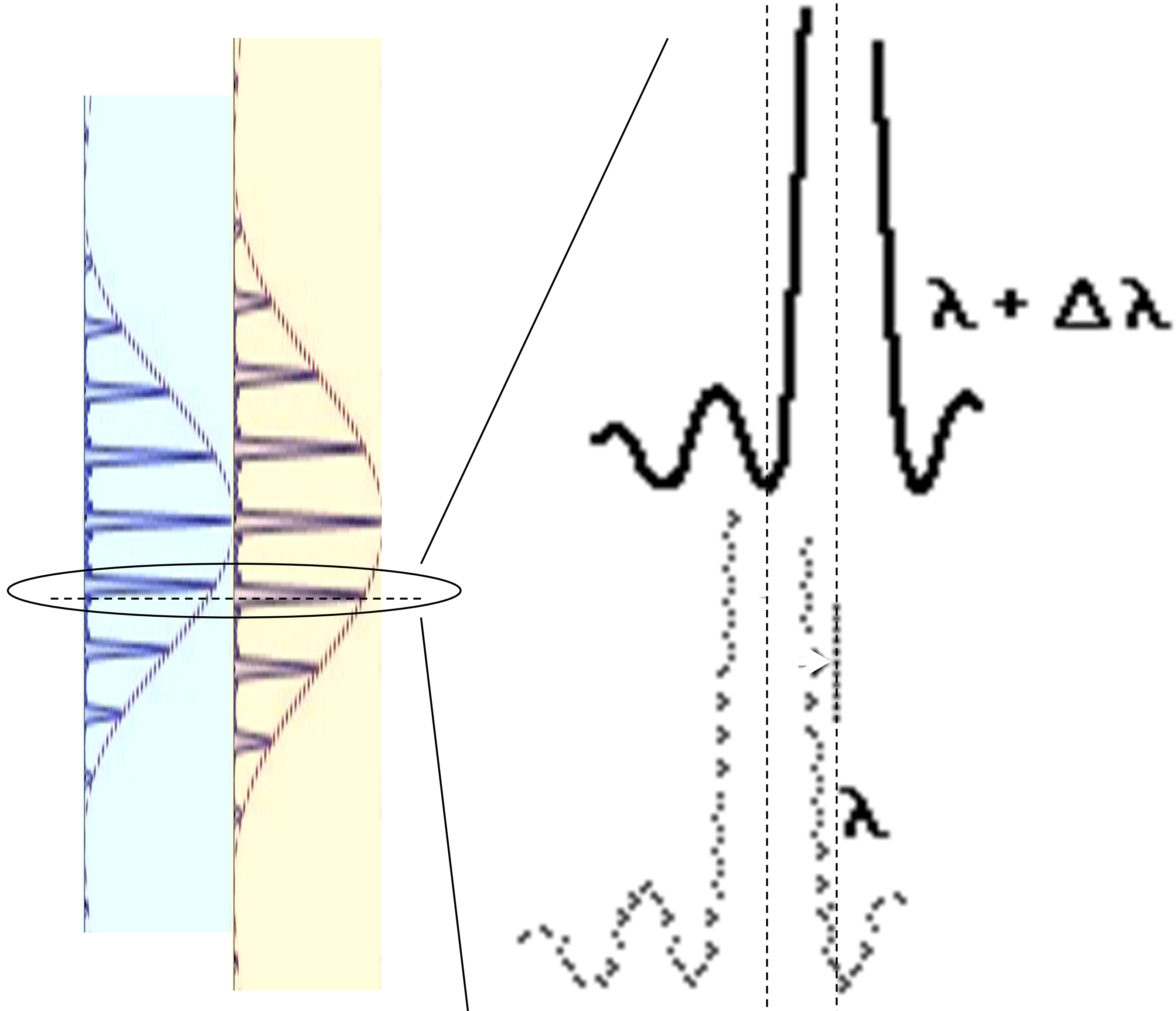


$$\phi \equiv kd\sin\theta$$
$$= \frac{2\pi}{\lambda} d\sin\theta$$

is phase shift between electric field contributions from neighboring slits

Resolving power of a diffraction grating:

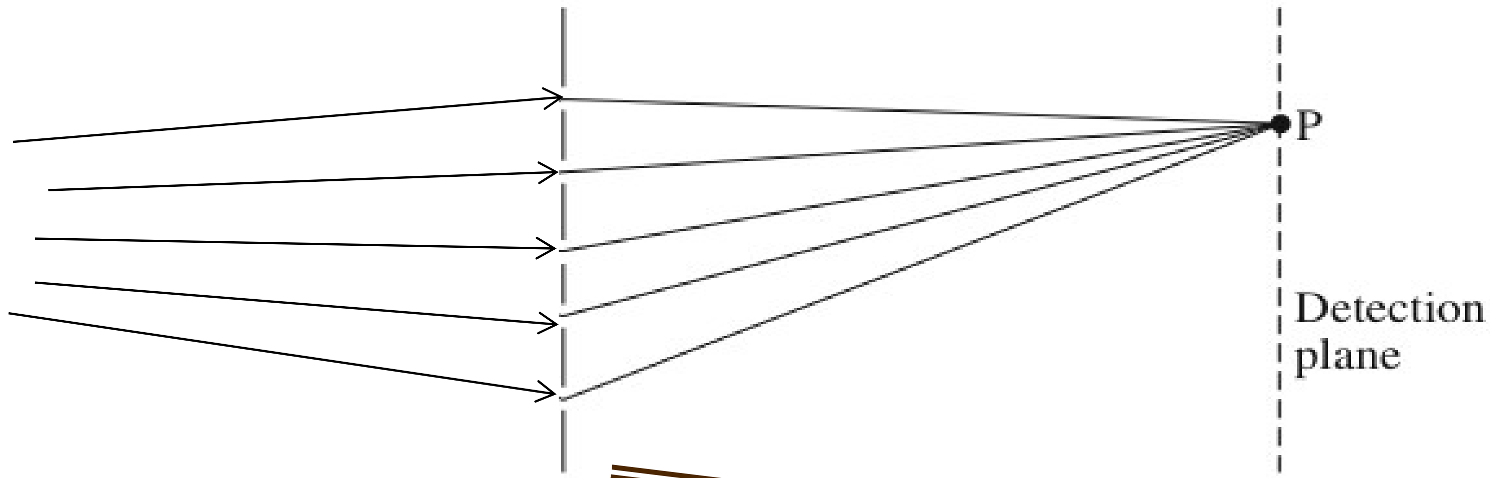
Light diffracts off N slits



$$\boxed{\mathcal{R} \equiv \frac{\lambda}{\Delta\lambda} = N}$$

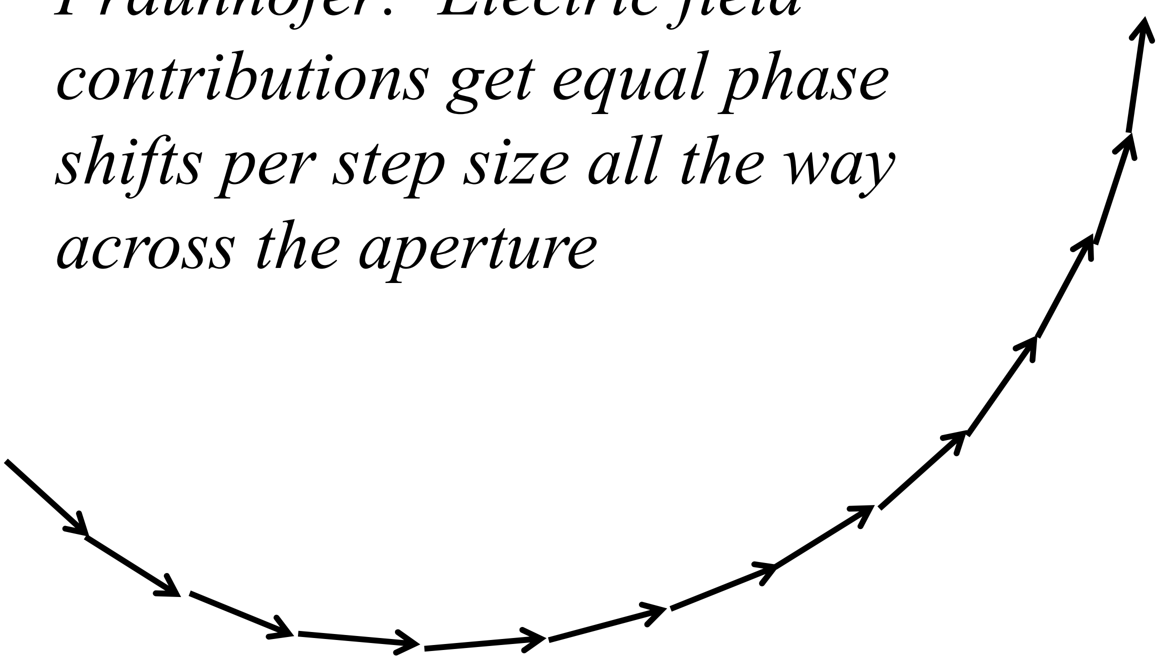
λ and $\lambda + \Delta\lambda$ just resolvable

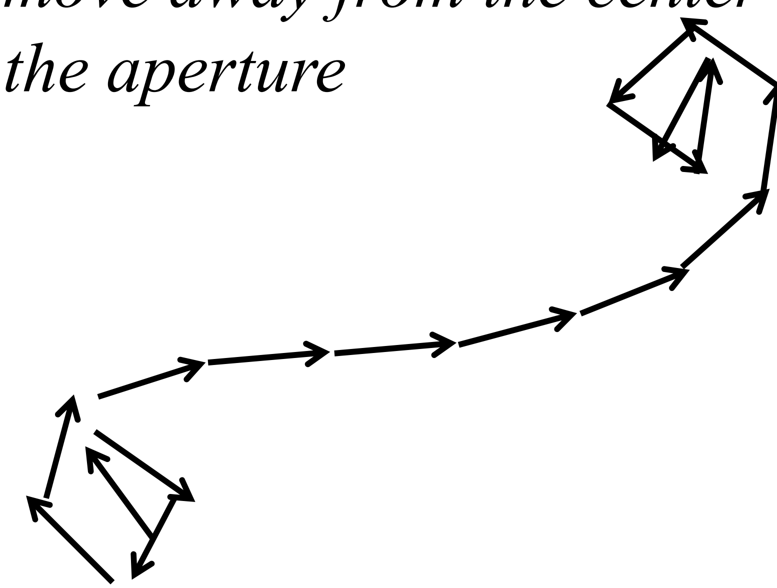
Fraunhofer vs. Fresnel diffraction?



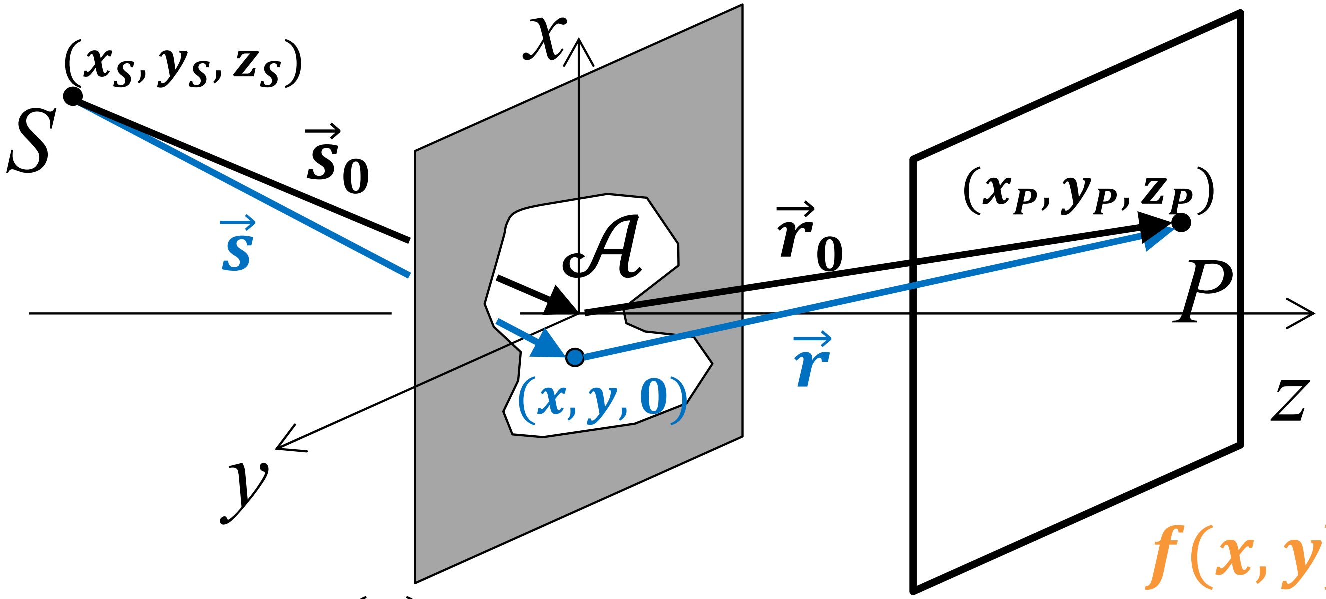
Fraunhofer: Electric field contributions get equal phase shifts per step size all the way across the aperture

Fresnel: Electric field contributions get larger phase shifts per step size as you move away from the center of the aperture





Fresnel Diffraction: setting up the math



$$f(x,y) \equiv (\mathbf{s} - \mathbf{s_0} + \mathbf{r} - \mathbf{r_0})$$

$$u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)}}{\lambda} \frac{e^{iks_0}}{s_0} \frac{e^{ikr_0}}{r_0} \frac{[\hat{s}_0 \cdot \hat{z} + \hat{r}_0 \cdot \hat{z}]}{2} \iint_{\mathcal{A}} dx\, dy\, e^{ik(s-s_0+r-r_0)}$$

$$f(x,y) \approx -\left(\frac{x_S}{s_0} + \frac{x_P}{r_0}\right)x - \left(\frac{y_S}{s_0} + \frac{y_P}{r_0}\right)y$$

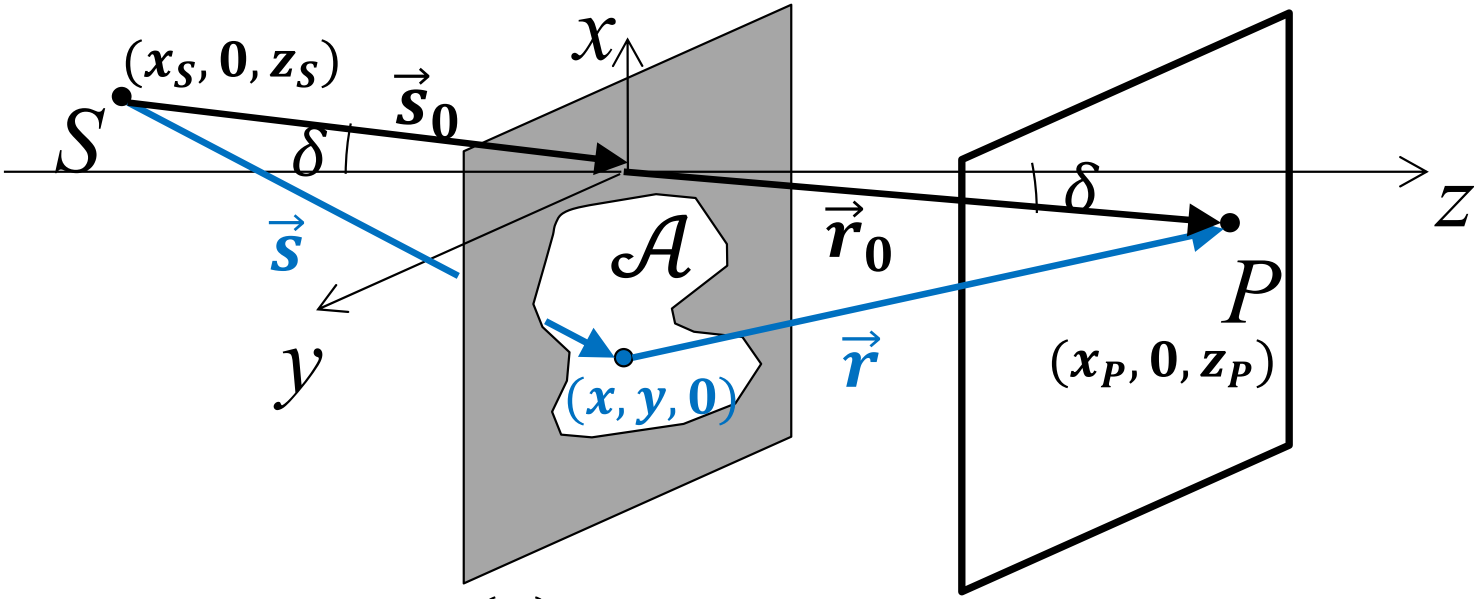
$$+\frac{1}{2}\left(\frac{1}{s_0} + \frac{1}{r_0}\right)(x^2 + y^2) - \frac{1}{2s_0}\left(\frac{x_S}{s_0}x + \frac{y_S}{s_0}y\right)^2 - \frac{1}{2r_0}\left(\frac{x_P}{r_0}x + \frac{y_P}{r_0}y\right)^2$$

Choose origin at intersection of $\overline{SP}$ with aperture plane.

Set y axis $\perp \overline{SP}$.

Thus S at $(\boldsymbol{x_S}, \mathbf{0}, \boldsymbol{z_S})$ and P at $(\boldsymbol{x_P}, \mathbf{0}, \boldsymbol{z_P})$.

Fresnel Diffraction: setting up the math II

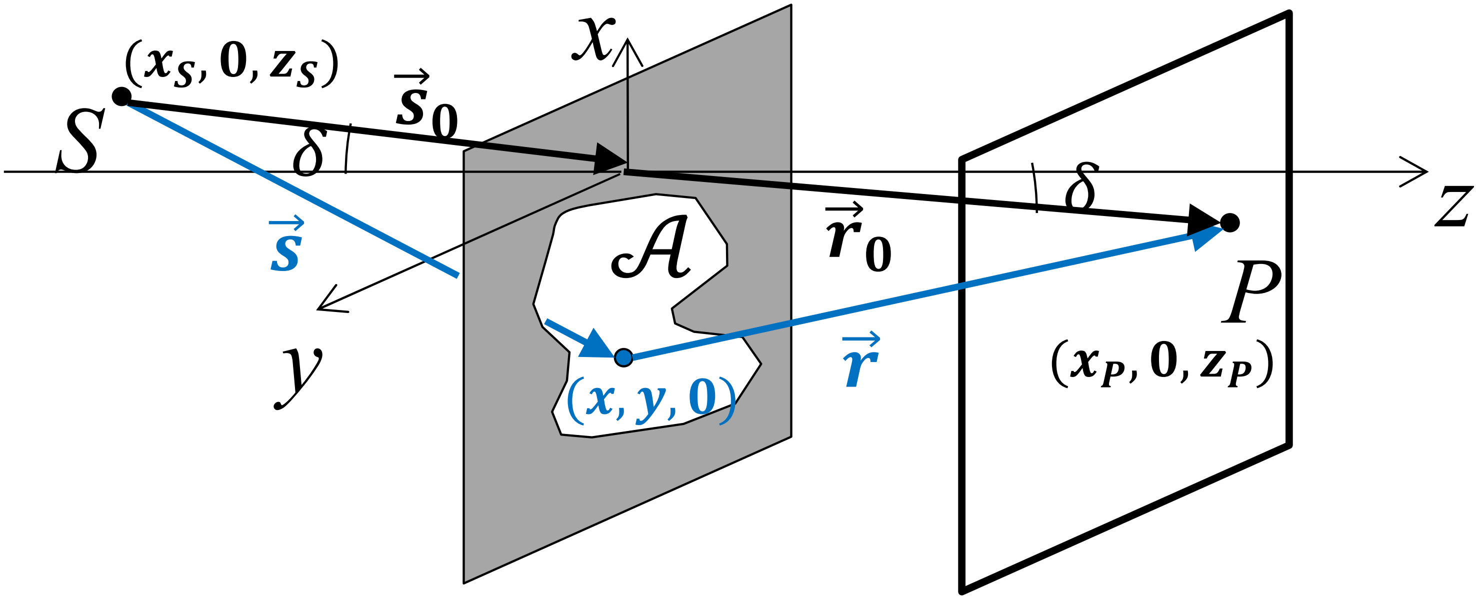


$$u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)}}{\lambda} \frac{e^{iks_0}}{s_0} \frac{e^{ikr_0}}{r_0} \frac{[\hat{s}_0 \cdot \hat{z} + \hat{r}_0 \cdot \hat{z}]}{2} \iint_{\mathcal{A}} dx\, dy\, e^{ikf(x,y)}$$

$$f(x,y) \approx -\left(\frac{x_S}{s_0} + \frac{x_P}{r_0}\right) x - \left(\frac{y_S}{s_0} + \frac{y_P}{r_0}\right) y$$
$$+ \frac{1}{2}\left(\frac{1}{s_0} + \frac{1}{r_0}\right)(x^2 + y^2) - \frac{1}{2s_0}\left(\frac{x_S}{s_0} x + \frac{y_S}{s_0} y\right)^2 - \frac{1}{2r_0}\left(\frac{x_P}{r_0} x + \frac{y_P}{r_0} y\right)^2$$

$$f(x,y) = \frac{1}{2}\left(\frac{1}{s_0} + \frac{1}{r_0}\right)(x^2 \cos^2\delta + y^2)$$

Fresnel Diffraction: setting up the math III

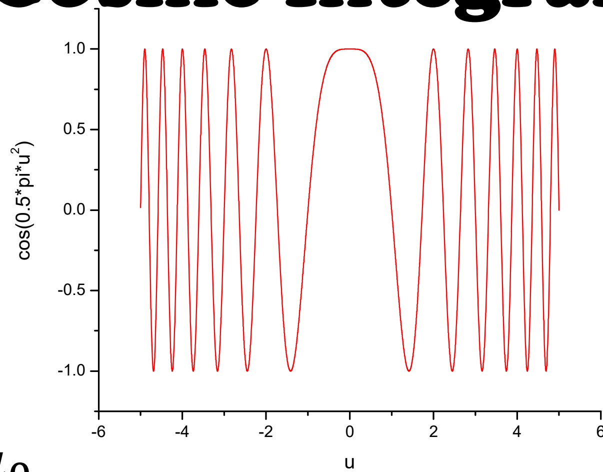


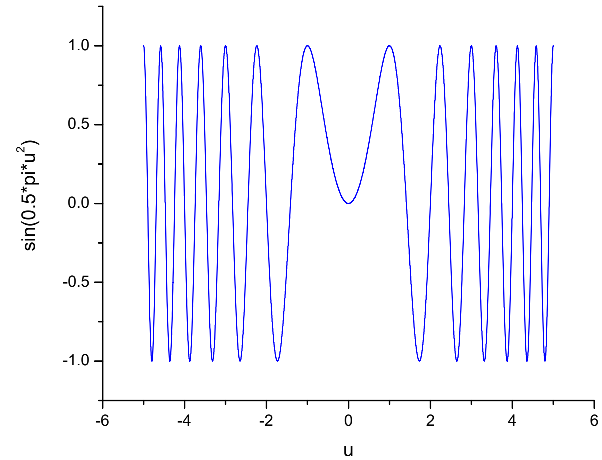
$$u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)}}{\lambda} \frac{e^{ik(s_0+r_0)}}{s_0 r_0} \cos\delta \iint_{\mathcal{A}} dx\, dy\, e^{ik\left[\frac{1}{2}\left(\frac{1}{s_0}+\frac{1}{r_0}\right)(x^2\cos^2\delta + y^2)\right]}$$

$$k\frac{1}{2}\left(\frac{1}{s_0}+\frac{1}{r_0}\right)x^2\cos^2\delta \equiv \frac{\pi}{2}u^2 \qquad k\frac{1}{2}\left(\frac{1}{s_0}+\frac{1}{r_0}\right)y^2 \equiv \frac{\pi}{2}\mathrm{v}^2$$

$$\boxed{u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)}}{2} \frac{e^{ik(s_0+r_0)}}{s_0 + r_0} \iint_{\mathcal{A}} du\, d\mathrm{v}\, e^{i\frac{\pi}{2}u^2} e^{i\frac{\pi}{2}\mathrm{v}^2}}$$

Fresnel Sine and Cosine Integrals





$$\int_0^{u_0} du\; e^{i\frac{\pi}{2}u^2} = \underbrace{\int_0^{u_0} du\;\cos\left(\tfrac{\pi}{2}u^2\right)}_{\substack{\equiv C(u_0) \\ \textit{Fresnel cosine integral}}} + \; i \underbrace{\int_0^{u_0} du\;\sin\left(\tfrac{\pi}{2}u^2\right)}_{\substack{\equiv S(u_0) \\ \textit{Fresnel sine integral}}}$$

$$C(-u_0) = -C(u_0)$$

$$S(-u_0) = -S(u_0)$$

$$\int_0^{\infty} du\,\cos\left(\tfrac{\pi}{2}u^2\right) = \frac{1}{2} = \int_{-\infty}^{0} du\,\cos\left(\tfrac{\pi}{2}u^2\right)$$

$$\int_0^{\infty} du\,\sin\left(\tfrac{\pi}{2}u^2\right) = \frac{1}{2} = \int_{-\infty}^{0} du\,\sin\left(\tfrac{\pi}{2}u^2\right)$$

$$\Longrightarrow \int_{-\infty}^{\infty} du\; e^{i\frac{\pi}{2}u^2} = 1 + i$$

The Cornu Spiral